

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 14: AC Generation, Alternators & CSD / IDG

Syllabus Source: CAE Oxford (pp. 97-138 (Elec)) | Pass Target: >= 90% AviationExam

1. AERONAUTICAL AC SPECIFICATIONS: 115V / 200V AT 400 HZ

Transport aircraft utilize 3-phase Alternating Current (AC) as the primary electrical generation medium due to dramatic weight savings and superior motor/transformer efficiency compared to DC systems:

- Voltage Specification: Standard commercial aircraft AC power is 115 Volts RMS Phase-to-Neutral (Star configuration) and 200 Volts RMS Line-to-Line (Phase-to-Phase):
 $V_{line} = \sqrt{3} \times V_{phase} = 1.732 \times 115V = 200V$.
- Frequency Specification (400 Hz): Commercial aircraft operate at 400 Hertz (cycles per second), compared to 50/60 Hz domestic ground mains. High 400 Hz frequency allows transformers, alternators, and motors to be manufactured with dramatically smaller, lighter magnetic iron cores, saving tons of aircraft weight!
- Frequency Formula: Frequency (f) = (P x N) / 120, where P = number of magnetic poles and N = generator rotational speed in RPM. For an 8-pole generator to produce 400 Hz: N = (120 x 400) / 8 = 6,000 RPM (CONSTANT)!

2. MODERN BRUSHLESS AC GENERATOR ARCHITECTURE

High-altitude sparking and rapid carbon brush wear in thin air led to the universal adoption of brushless AC generators, housing three distinct electrical machines on a single rotating rotor shaft:

Stage	Rotating Rotor Element	Stationary Stator Element & Function
1. Pilot Exciter	Permanent Magnet Rotor (spinning magnets).	Permanent Magnet Generator (PMG) 3-phase stator. Induces AC voltage to power the Generator Control Unit (GCU) independently without battery power!
2. Main Exciter	3-phase AC armature winding feeding rotating rectifier diodes.	DC Field Winding controlled by the GCU. GCU varies DC current to regulate main output voltage.
3. Rotating Rectifier	Six silicon diodes mounted on the rotating shaft.	Rectifies 3-phase AC from main exciter rotor into DC, directly feeding the main generator field WITHOUT BRUSHES!
4. Main Generator	Heavy DC magnetic field windings (poles).	Main 3-phase Star-connected stator. Delivers primary 115V / 200V 400 Hz power to the main aircraft busbars.

3. CONSTANT SPEED DRIVE (CSD) & INTEGRATED DRIVE GENERATOR (IDG)

Because jet engine accessory gearbox speed varies constantly (from ~4,500 to ~9,000 RPM), an intermediate hydromechanical transmission is mandatory to drive the generator at an EXACT, UNVARYING 6,000 RPM to maintain 400 Hz (+/- 1% / 396 to 404 Hz):

- Constant Speed Drive (CSD): A variable hydromechanical transmission (hydraulic pump and motor linked through a differential planetary gear train) that adds or subtracts speed to keep the output shaft locked at 6,000 RPM.

- Integrated Drive Generator (IDG): The modern standard (e.g. A320, B737NG, B777). Integrates the CSD transmission and the brushless AC generator inside a single, common oil-cooled casing, saving weight and maintenance.
- CSD / IDG In-Flight Disconnect: If CSD oil temperature exceeds limits or oil pressure drops, a cockpit DISCONNECT switch fires an electro-mechanical solenoid that physically disengages a dog-clutch from the engine gearbox. CRITICAL EXAM POINT: Once disconnected in flight, the CSD / IDG CANNOT BE RECONNECTED IN FLIGHT! It can ONLY be reset manually by ground mechanics after landing!

[!] EXAM TRAP (AVIATIONEXAM) - CSD / IDG Disconnect Rules & Warnings

- Cockpit Disconnect Switch: Protected by a red guarded cover. Activated when high oil temperature or low oil pressure is annunciated. - Thermal Disconnect: Many IDGs incorporate a eutectic solder plug that automatically disconnects the drive if case temperature exceeds ~180°C. - STRICT EXAM RULE: The CSD CANNOT be reconnected in flight under any circumstances!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): IDG In-Flight Disconnect Protocol

Scenario: IDG Oil Temp amber alert or low drive oil pressure. Step 1: Verify associated engine is RUNNING (rotation is required to separate the spiral dog clutch). Step 2: Lift guard and depress IDG DISCONNECT switch for 3 seconds. Step 3: The drive disconnects mechanically via spring-loaded solenoid. Step 4: Crucial Exam Rule: The IDG CANNOT be reconnected in flight under any circumstances. Reset is done on ground only.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why do commercial transport aircraft use an electrical AC frequency of 400 Hz instead of standard commercial power?

- [A]: To prevent radio interference
- [B]: Because 400 Hz allows transformers, motors, and magnetic components to be engineered with much smaller and lighter iron cores, saving substantial aircraft weight
- [C]: Because 400 Hz does not cause electrical shocks
- [D]: To improve generator brush life

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The size and weight of electromagnetic iron cores (in transformers, generators, and electric motors) is inversely proportional to frequency. 400 Hz allows miniature, lightweight components compared to bulky 50/60 Hz industrial equipment.

Q2: In a 3-phase 115V AC star-connected aircraft generator, what is the line-to-line (phase-to-phase) voltage?

- [A]: 115 V
- [B]: 200 V
- [C]: 230 V
- [D]: 345 V

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Line-to-line voltage is related to phase-to-neutral voltage by $V_{line} = \sqrt{3} \times V_{phase}$. Thus: $1.732 \times 115V = 199.2 V$ (~200 Volts AC).

Q3: Can an Integrated Drive Generator (IDG) that has been disconnected in flight by the pilot be reconnected b

- [A]: Yes, by pressing the IDG reset button on the overhead panel
- [B]: Yes, once the oil cools below 100°C
- [C]: No, mechanical reconnection is physically impossible in flight and can only be performed manually on the ground by maintenance personnel
- [D]: Yes, by recycling the engine master switch

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. The CSD/IDG disconnect separates a mechanical spring-loaded dog clutch. Once pulled out of mesh, it cannot be engaged while running and must be reset manually on the ground.